Naby Diop

DATA SCIENTIST | STATISTICAL MODELING | RISK ANALYTICS

PROFESSIONAL SUMMARY (Contact Me (Portfolio Website))

Data Science & Strategy Leader with a Master's in Statistics and extensive expertise in transforming complex banking data into institutional-grade assets. Advanced specialist in engineering scalable Python (Polars/Pandas), R, and SQL pipelines to automate ETL workflows, validate data quality, and calibrate credit-risk models. Highly adept at translating complex methodologies into compliant Global Emerging Markets (GEMs) submissions and custom web applications (Flask) while bridging Data Engineering, Risk, and Finance to drive executive-level decisions.

CORE SKILLS

Python | R | SQL | Pandas | Polars | Flask | Statistical Modeling | Machine Learning | Survival Analysis | Time Series | Feature Engineering | Data Cleaning | Data Visualization | Shiny | Streamlit | Git

PROFESSIONAL EXPERIENCE

ELMA GROUP: Head of Strategy & Financing – Data Strategy Lead | 2024-Present

- Drive strategic financial planning and digital content management at ELMA Group to enhance operational effectiveness.
- Developed a custom Flask application for a public website, enabling independent content management.
- Created a centralized platform to showcase ELMA's publishing activities and pan-African identity, boosting our digital presence.

Freelance: Outlier, DataAnnotation, Mercor, Upwork | 2023-Present

- Engineered End-to-End ETL Pipelines: Architected scalable Python (Polars/Pandas) and SQL data pipelines to ingest, clean, and standardize heterogeneous data sources into analysis-ready datasets.
- Automated GEMs Reporting Solutions: Designed and deployed an automated data extraction and validation architecture tailored strictly to Global Emerging Markets (GEMs) structures and reporting standards.
- Led Cross-Functional Data Governance: Collaborated with Risk Management, Finance, IT, and Operations to document auditable workflows and produce reliable datasets for portfolio and credit-risk analysis.

SELECTED DATA SCIENCE PROJECTS

Independent Data Science Projects | World Bank & FAO Databases | 2020-2023

- Developed an interactive application for analyzing agricultural production data across various countries and crops.
- Processed over 216,000 FAOSTAT records from 2000 to 2021 using R and Tidyverse.
- Created a Shiny dashboard featuring Plotly maps and linear-trend forecasts for enhanced user experience.
- Designed an interactive survival analysis application for colorectal cancer patient outcomes, making statistical results accessible.

EDUCATION & TRAINING

M.A. & B.A., Statistics / Applied Mathematics - CUNY Hunter College | 01/2017 - 06/2020
B.S., Mathematics - University Cheikh Anta Diop (UCAD) | 10/2011 - 06/2015
Certificate, Business Data Science - Business Science University | 01/2020 - Present
Certificate, Basic Credit Risk Modeling for Basel/IFRS 9 using R/Python/SAS, BlueCourses Certificate

CONTACT

Email: naby.diop07@yahoo.com | Location: New York, United States